

TSCINBUNY

ESP32 CAM McKnum Wheel ROBOT V1.0

V2.02.505.08

Tutorial Introduction

This tutorial includes the following files to help you learn more about the car kit and programming knowledge:



"1_Get_start": This folder contains the robot assembly guide and necessary software environment files. In order to complete the assembly accurately and quickly, please read and follow the manual in detail. At the same time, the premise of realizing the program function is to correctly create the software environment. Please check the PDF files in this folder first .

"2_Arduino_Code": This folder is used to store Arduino code files. Each code file is uploaded and used independently.

“3_References”: This folder stores some reference materials;

After completing the assembly of the car and creating the programming environment, follow this tutorial step by step to complete the burning of the smart car program and realize different functions!

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1. Upload UNO code

1.1 Description

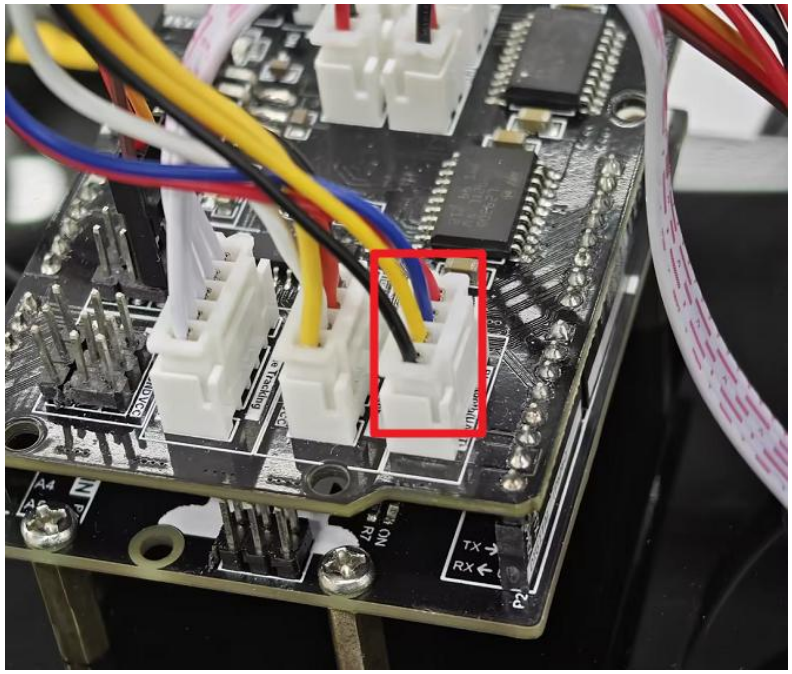
This section mainly introduces the use of TSCINBUNY control board and how to upload UNO code.

1.2 Upload code

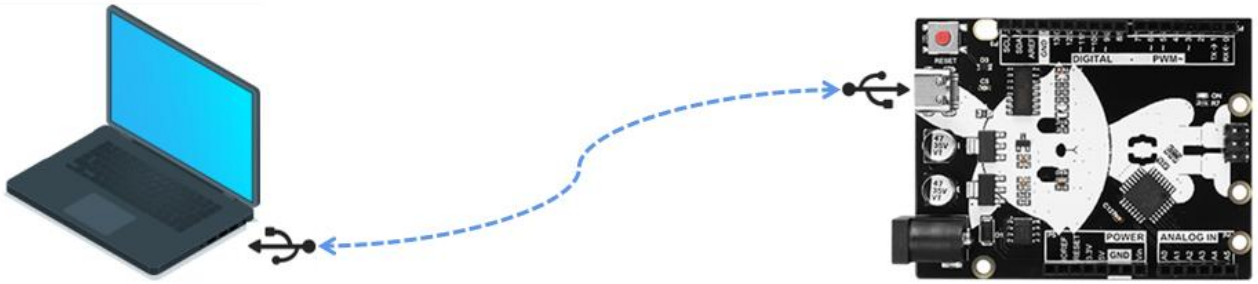
Open the code file (path: 2_Arduino_Code\6.2_Arduino_UNO\6.2_Arduino_UNO.ino)



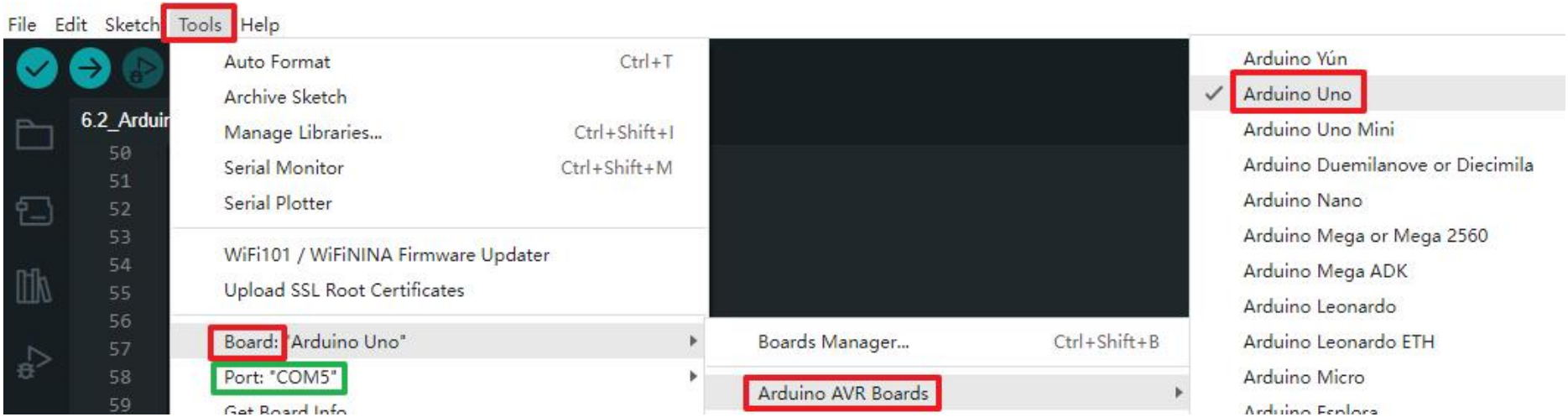
unplug the four wires VCC/GND/RX/TX before uploading the program .



Connect the Arduino board to your computer using a USB cable .

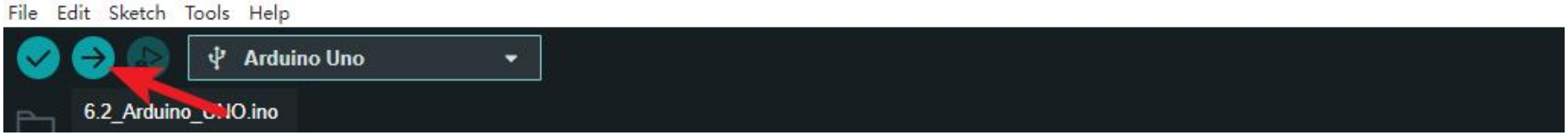


Select Uno for board type and COM5 for serial port

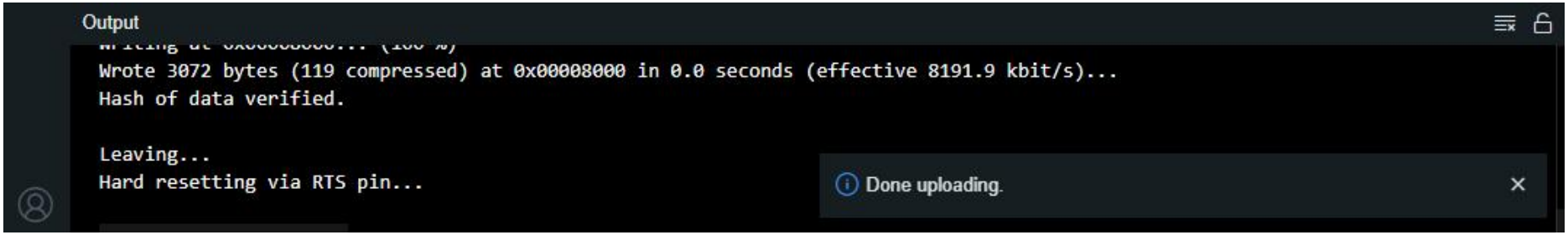


Note: In reality everyone's serial port display will be different, although COM 5 is selected here , it may be COM3 or COM4 on your computer.

Click the "Upload" button and the program starts uploading.



After the upload is successful, the message "Done uploading" will be displayed.


Output
Wrote 3072 bytes (119 compressed) at 0x00008000 in 0.0 seconds (effective 8191.9 kbit/s)...
Hash of data verified.
Leaving...
Hard resetting via RTS pin...
Done uploading.

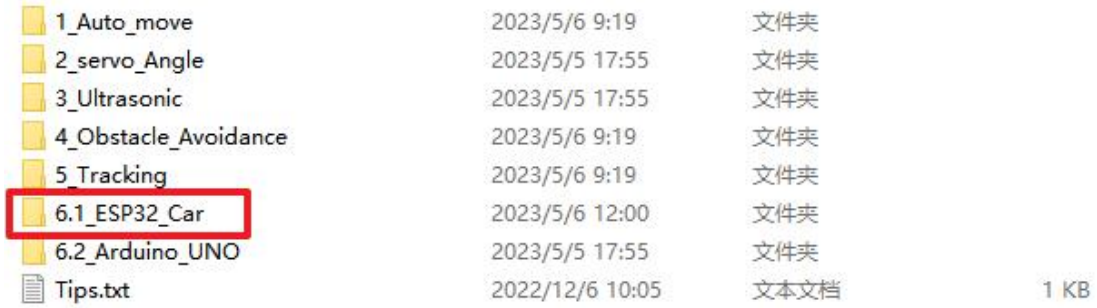
2.ESP32 CAM

2.1 Description

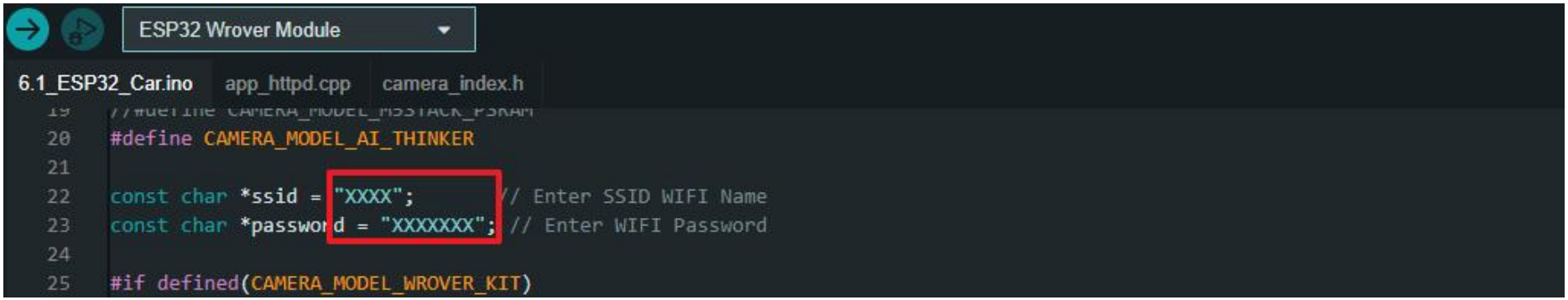
This section mainly introduces the use of ESP32 CAM, learns how to upload code and realize the display of the captured image in real time on the browser after ESP32 CAM is connected to WiFi.

2.2 Upload code

Open the code file (path: 2_Arduino_Code\6.1_ESP32_Car\6.1_ESP32_Car.ino)



network credentials account and password variables in the code to your own WiFi name and password !

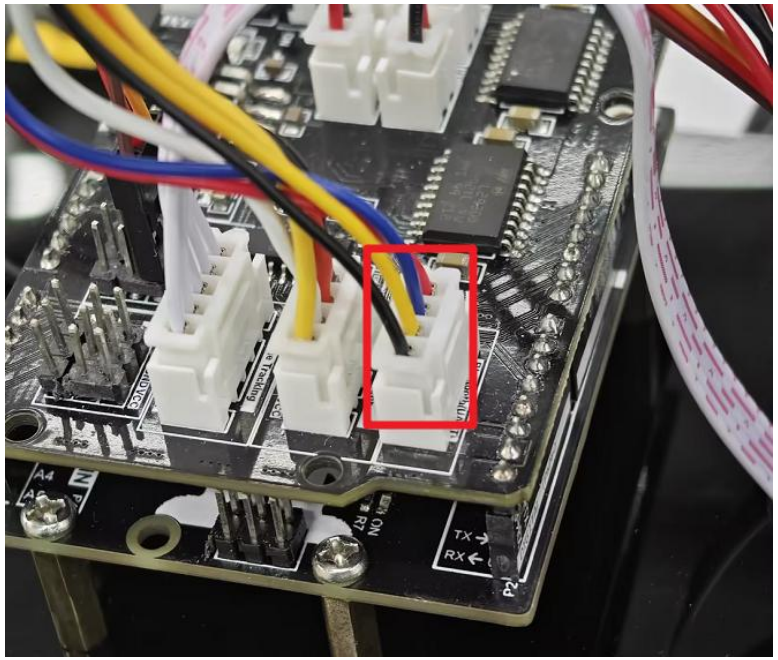


ssid = " Your wifi name " ;

password = " WIFI Password " ;

At the same time, you should check that the WiFi you are connecting to is 2.4G instead of 5G band, otherwise the function cannot be realized!

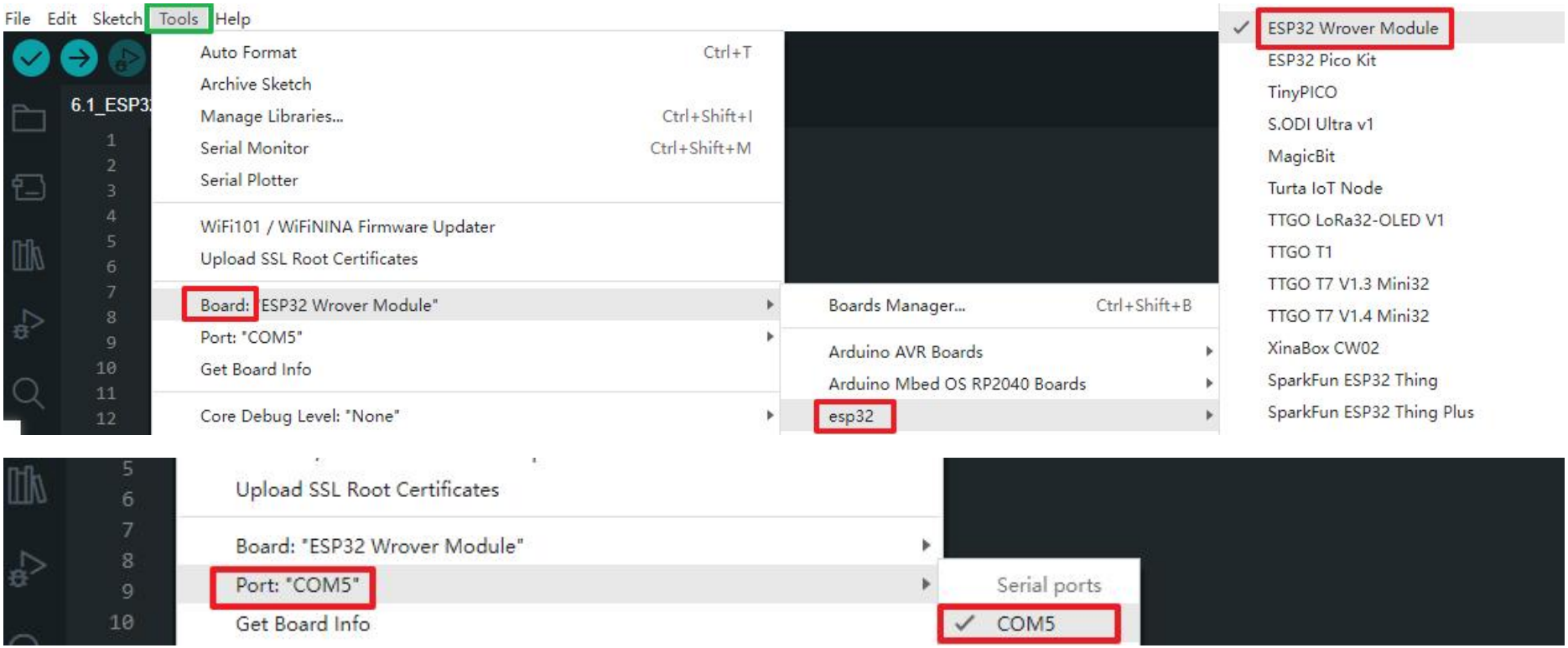
unplug the four wires VCC/GND/RX/TX before uploading the program .



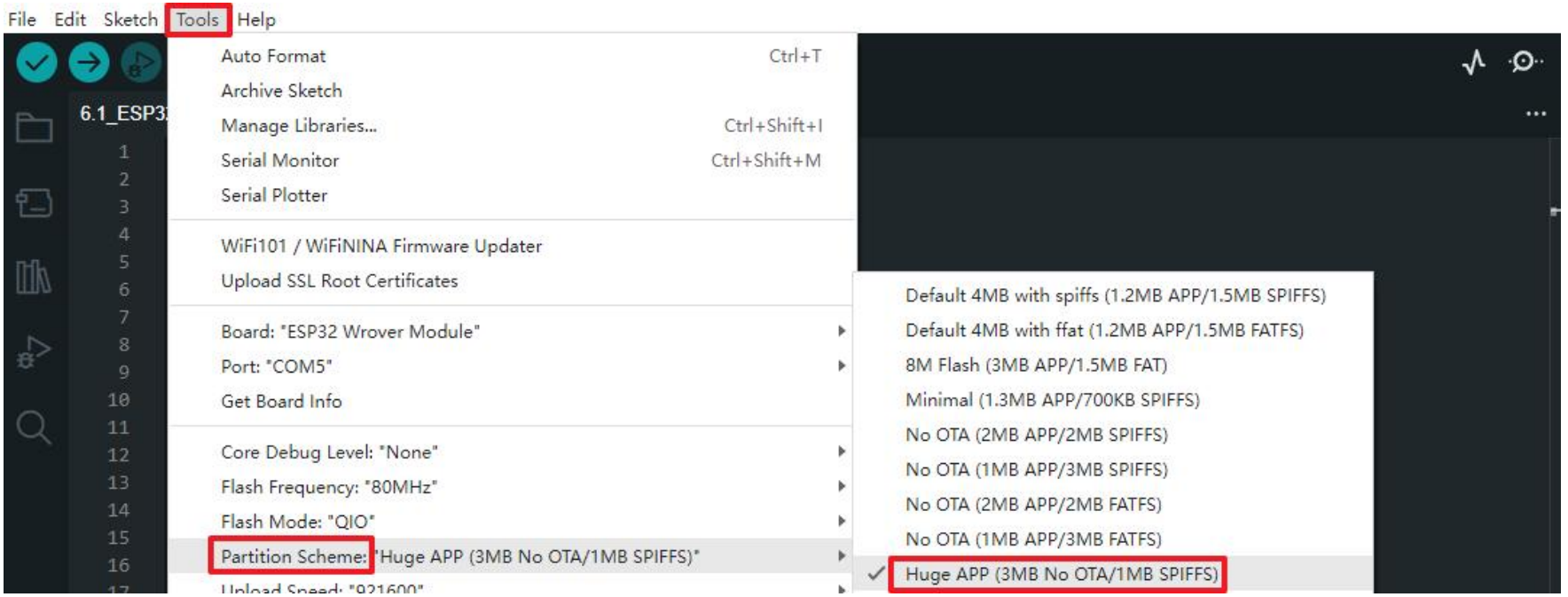
Connect the ESP32 CAM board to your computer with a USB cable



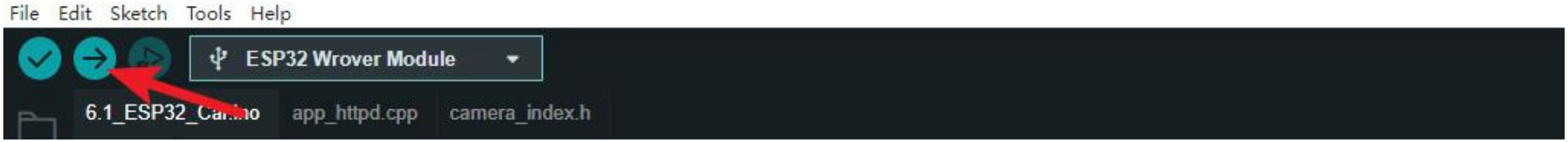
Select the board type as ESP32 and the serial port number as COM5



Note: Select ESP32 Wrover as the board type and COM5 as the serial port. In fact, the serial port displayed by each person is different. Although COM 5 is selected here, it may be COM3 or COM4 on your computer.



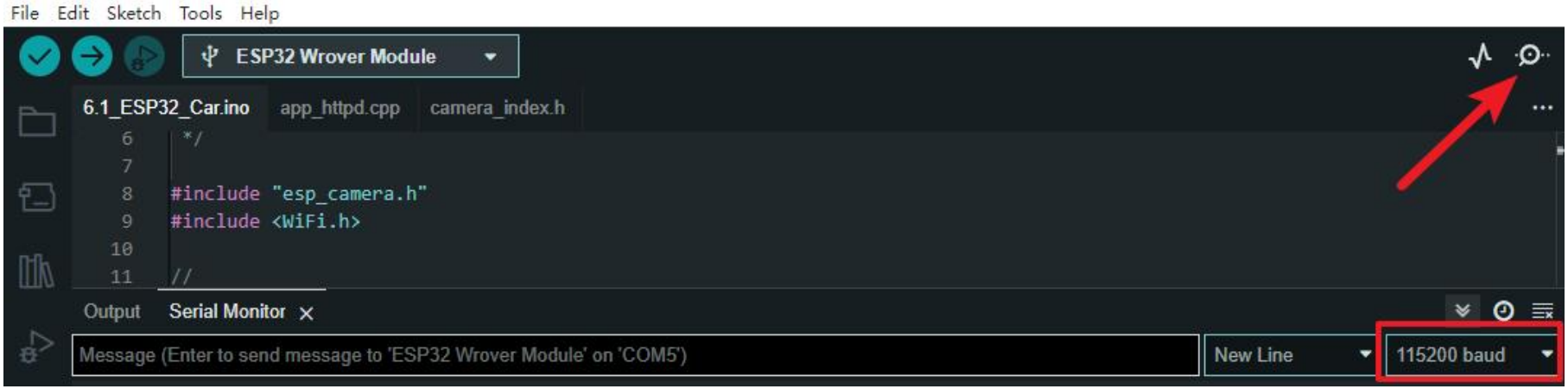
Click the "Upload" button and the program starts uploading.



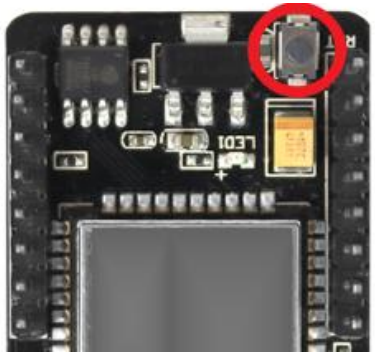
After the upload is successful, the message "Done uploading" will be displayed.

2.3 Check IP address

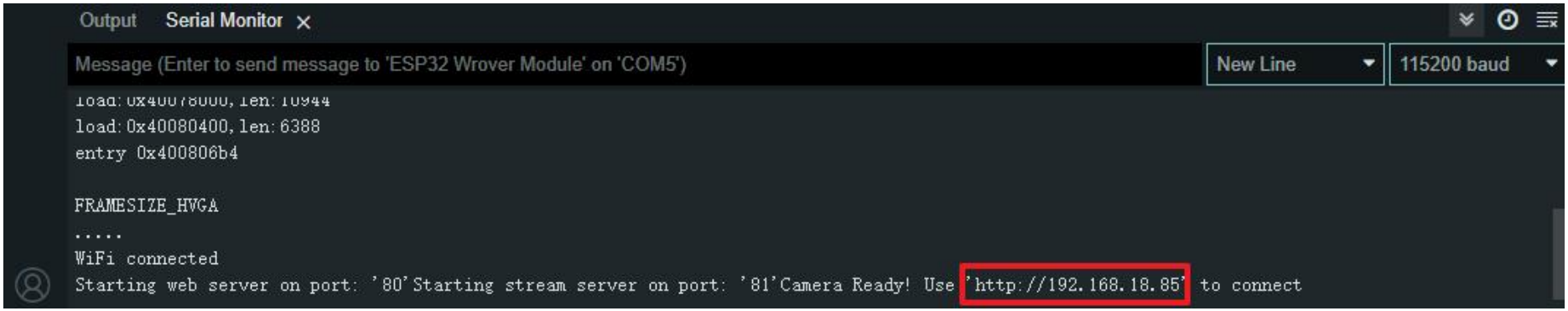
Open the serial monitor and set the baud rate to 115200 in the lower right corner



Press the reset button on the ESP32 CAM to reload the program.(Or disconnect the USB and plug it back in to reset)



When the program reloads and successfully connects to WiFi, the IP address will be displayed: http://xxx.xxx.xxx.xxx, please write it down



3. APP remote control car

3.1 Android devices

Open the directory folder 4_APP and install "TSCINBUNY.apk" on your phone. When you see this tutorial, there may be a newer version of the software. When prompted to upgrade, please allow the upgrade and keep the phone connected to the network.



3.2 iOS devices

Open the App Store, search for and install TSCINBUNY

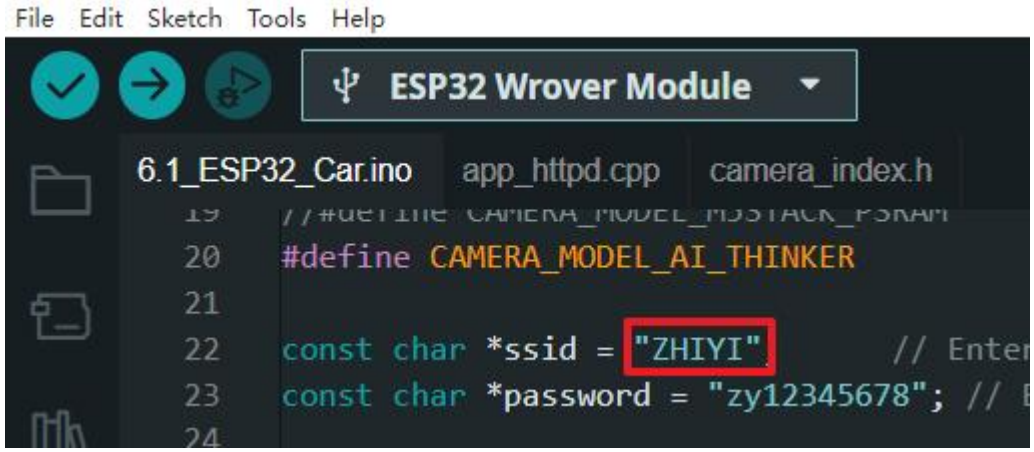


3.3. APP and ESP32 communication

Next , we will take an Android phone as an example to demonstrate how to control a smart car through this application :

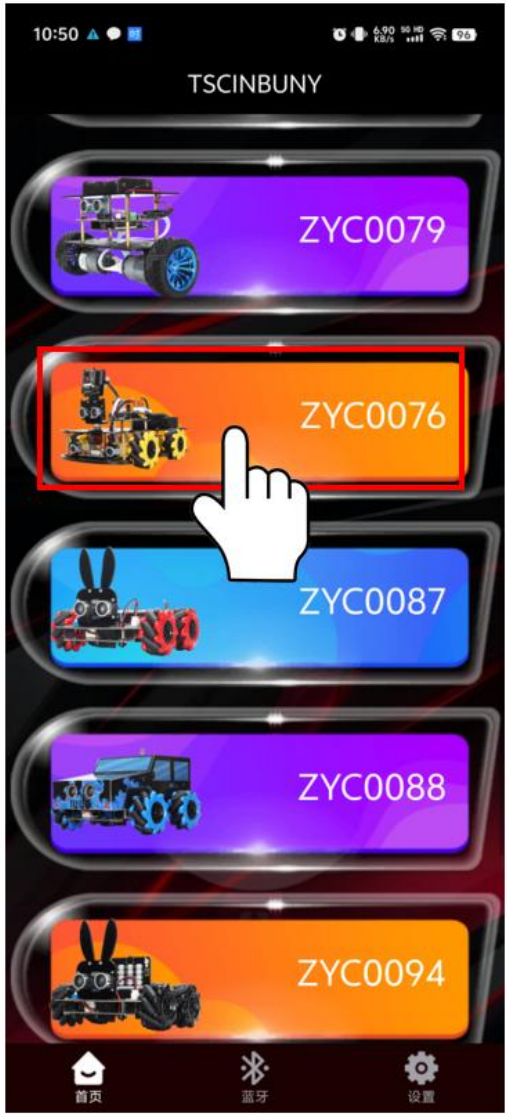
turn on the phone's WIFI switch and connect to the corresponding WiFi in the code



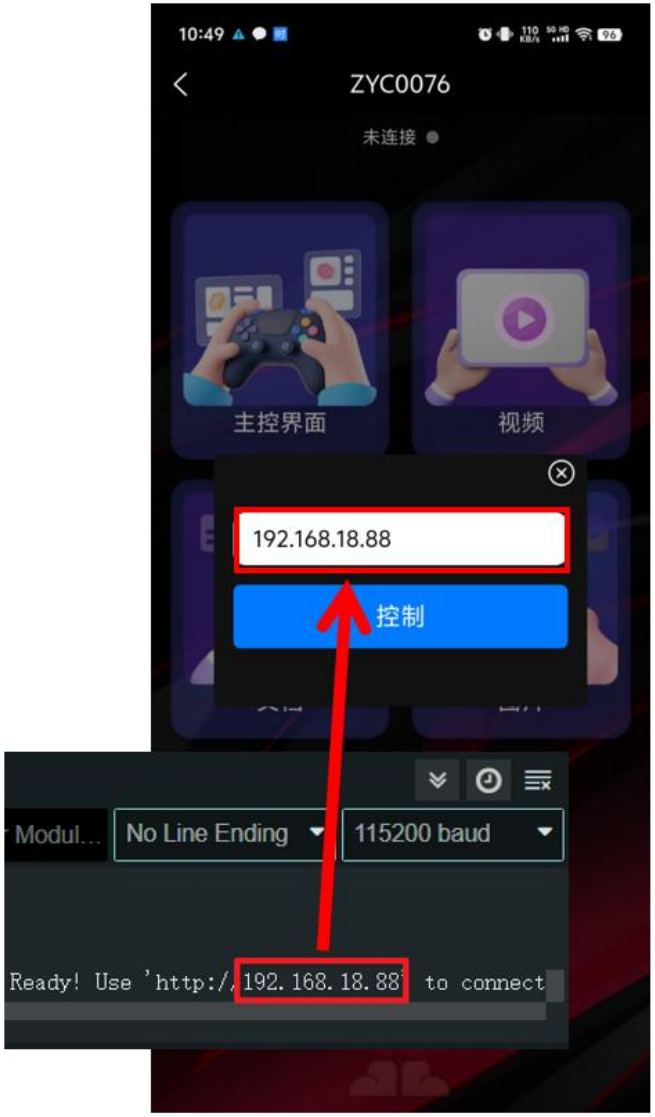


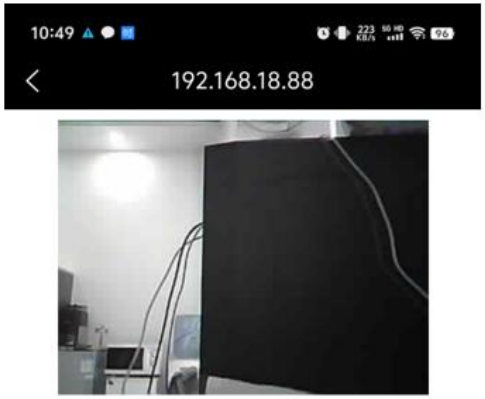
Then open the newly installed TSCINBUNY remote control APP, select the purchased product SKU> enter the main interface> fill in the IP address printed by the Arduino IDE serial port monitor into the input box and click OK to control.

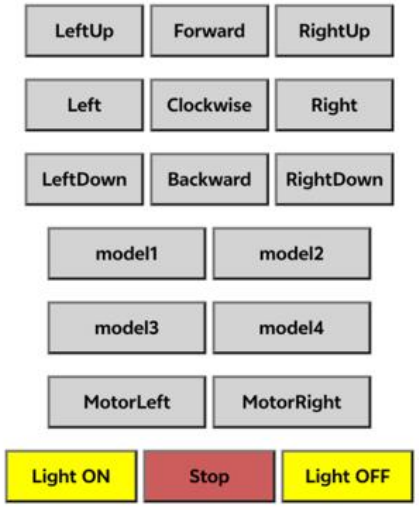
(The IP address will change after changing the network, and the IP obtained after uploading the code will prevail)

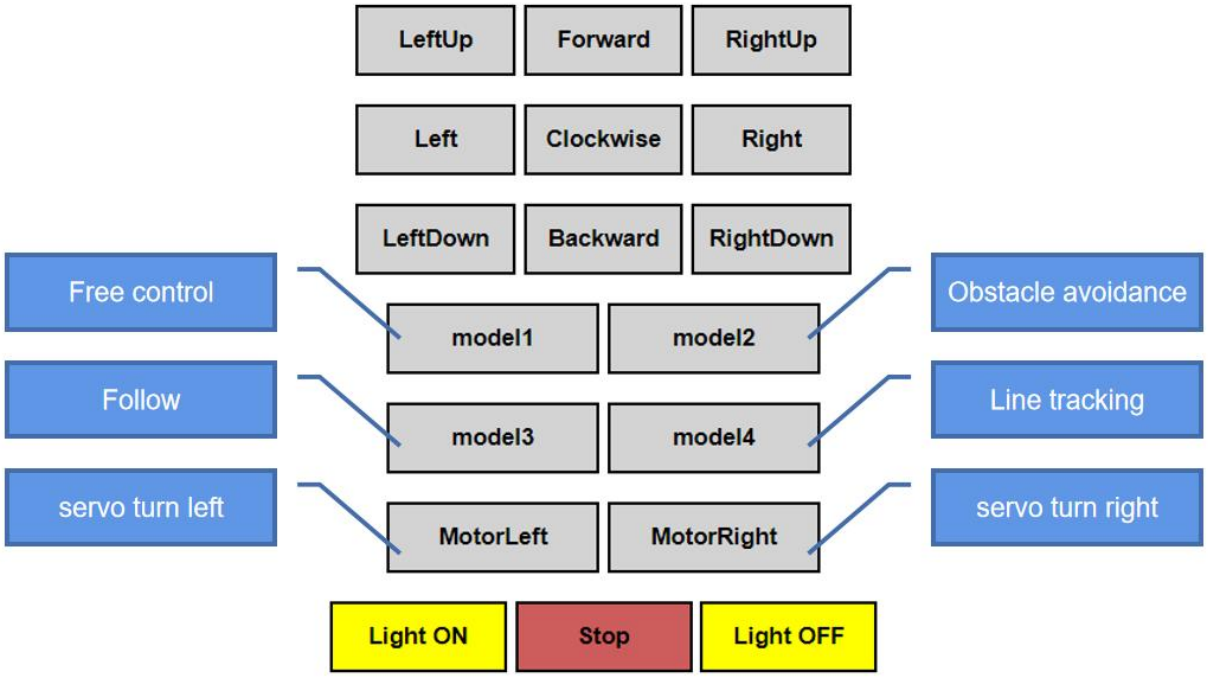

10:50
TSCINBUNY
ZYC0079
ZYC0076
ZYC0087
ZYC0088
ZYC0094
首页
蓝牙
设置


10:50
ZYC0076
未连接
主界面
视频
文档
图片


10:49
ZYC0076
未连接
主控界面
视频
192.168.18.88
控制
No Line Ending
115200 baud
Ready! Use 'http://192.168.18.88' to connect







Tips: If the Android phone cannot be controlled in the APP, please switch to the browser and enter the same IP to control

the car.

When you can't see the picture:

1. Please check whether the WiFi name and password have been changed correctly;

2. The WiFi modified in the code is 2.4G frequency band, and the signal is good;

3. Reset ESP32 CAM to ensure stable power supply and sufficient battery power;

4. Check whether the network is unobstructed, stable and free of interference;

For more tutorial content, please see folder 3_Reference

